

Tales Imbiriba

Assitant Research Professor
Electrical and Computer Engineering
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EDUCATION

2012–2016 *Ph.D.*, Universidade Federal de Santa Catarina, Brazil, Dept of Electrical and Comp. Eng.
2006–2008 *M.Sc.*, Universidade Federal do Pará, Brazil, Dept of Electrical and Computer Engineering.
2001–2005 *B.Sc.*, Universidade Federal do Pará, Brazil, Dept of Electrical and Computer Engineering.

PROFESSIONAL EXPERIENCE

08/2023–Present **Assistant Research Professor**, Northeastern University, Electrical & Computer Engineering Dept, Boston, MA, USA. With joint appointment with the Institute for Experiential AI (EAI).

05/2021–07/2023 **Associate Research Scientist**, Northeastern University, Electrical & Computer Engineering Dept, Boston, MA, USA. Affiliated with the: Center for Signal Processing, Imaging, Reasoning, and Learning (SPIRAL)

2019–2020 **Lecturer**, Northeastern University, USA.
Classes: DS5010 Introduction to Data Management and Processing; DS5010 Introduction to Programming; EECE2300 Computational Methods for Data Analytics.

04/2019–05/2021 **Postdoctoral Research Associate**, Northeastern University, Electrical & Computer Engineering Dept, Boston, MA, USA. Affiliated with the: SPIRAL

03/2017–04/2019 **Postdoctoral Research Associate**, Universidade Federal de Santa Catarina, Electrical & Computer Engineering Dept, Florianópolis, SC, Brazil.

01/2016–12/2016 **Lecturer**, Instituto Federal de Santa Catarina, Florianópolis, SC, Brazil.
Classes: Digital Systems; Introduction to computer programming; C-programming; Advanced C-programming; Microcontrollers.

08/2008–03/2012 **Electrical Engineer**, FAPEU–Universidade Federal de Santa Catarina, Florianópolis, SC, Brazil. Activities included: Fixed-point DSP programming (Blackfin from Analog devices). Implementation of VoIP algorithms. Development of Echo cancellation systems using adaptive filtering. Delay estimation using wavelet transform.

AWARDS & DISTINCTIONS

1. 956 citations, h-index 16, i10-index 21 (source: [Google Scholar](#)).
2. Best Technical paper award 2023 IEEE/ION Position, Location and Navigation Symposium (PLANS).
3. Faculty Research Team Award, 23rd Annual Faculty and Staff Awards, COE, NU, Apr. 2021.
4. Best Technical Paper PETRA Award 2020.
5. Recipient of the Brazilian Ministry of Education Postdoctoral Scholarship, 2017.
6. Recipient of the Brazilian Ministry of Education Ph.D. Scholarship, 2012.
7. Recipient of the Brazilian Ministry of Education M.Sc. Scholarship, 2006.

R&D FUND RAISING

Grants Awarded as a Co-Principal Investigator

1. M. Fried-Oken, T. Imbiriba, T. Mommont, D. Erdogmus, RO1 Administrative Supplements to Support Collaborations to Improve the AI/ML-Readiness of NIH-Supported Data. Amount: \$200K. **Awarded August 2023.**
2. P. Closas, T. Imbiriba. *Jamming resilient satellite-aided wireless navigation*, NSF-Small-CIF, **Role: Co-PI**. Grant Amount: \$1.2M (Northeastern University split \$600K). **Awarded August 2023.**

3. M. Goodwin, T. Imbiriba, D. Erdoğan, *Enhancing Automated Prediction of Challenging Behavior in Individuals with Autism Using Biosensor Data and Machine Learning*. NSF-SCH (funded as a NIH-RO1), Co-PI, 2022–2025, Grant Amount: \$1.2M.
4. P. Closas, E. Beighley, D. Erdoğan, T. Imbiriba. *Spaceborne Multispectral Image Fusion for Water Mapping and Flood Management*, National Geographic Society, 2021. Grant Amount: \$98K.

Grants Awarded as Senior Personnel

1. Z. C. Danziger, G. Guidoboni, D. Erdoğan, J. Yin. A new hybrid modeling framework combining biophysics and deep learning to predict and optimize peripheral neuromodulation outcomes in lower urinary tract disease. 2022-2027. Grant Amount: \$3.6 M.
2. D. Erdoğan, S. Ostadabbas, T. Padi, M. Pavel. *Supporting Aging In Place Through Multimodal Sensing and Reasoning*. Verizon, Senior Personnel, 2021, \$650K.
3. D. Erdoğan, K. Chowdhury, P. Closas, Y. Wang, T. Melodia. *SPiNN for Wireless IoT*, DARPA, Senior Personnel, 2020. Grant Amount: \$1M.
4. Z. C. Danziger, E. Alhajjar, D. Erdoğan, G. Guidoboni. NIH, *A New Paradigm for Systems Physiology Modeling: Biomechanistic Learning Augmentation with Deep Differential Equation Representations (BLADDER)*, Senior Personnel, 2020. Grant Amount: \$2M.

Grants to be submitted

1. T. Imbiriba, E. Beighley, P. Closas. *Multimodal sensor fusion for algal bloom estimation in freshwater bodies*, NSF-HS. **Role: PI**. Grant Amount: \$400K. **Submission by Jan 31st.**
2. P. Closas, J. Dunik, T. Imbiriba, O. Straka. *Online Learning of Artificial Intelligence Enhanced Dynamical Physical Models*. NSF-GACR. **Role: Co-PI**. Grant Amount: \$600K. **re-Submission by December 31st.**
3. A. Stubbins, T. Imbiriba, A. Mueller, D. Erdoğan. *Computer-vision based macro-plastic flow estimation in river bodies using hyperspectral data*. NSF-HS. **Role: Co-PI**. Grant Amount: \$500K. **Submission by Feb 30.**

Grants submitted but not funded

1. P. Closas, T. Imbiriba. *Online Learning of Artificial Intelligence Enhanced Dynamical Physical Models*. NSF-GACR. **Role: Co-PI**. Grant Amount: \$600K. **Submitted March 1st.**
2. P. Closas, E. S. Lohan, H. Kuusniemi, T. Imbiriba. *Distributed AI for enhanced security in satellite-aided wireless navigation*. NSF-AoF. **Role: Co-PI**, \$600K
3. T. Imbiriba, E. Beighley, P. Closas, D. Erdoğan. NASA, *Multimodal sensor fusion for algal bloom estimation in water bodies*. **Role: PI**. Grant Amount: \$600K.
4. D. Erdoğan, A. R. Ganguly, M. Sznai, P. Closas, T. Imbiriba, O. Camps, J. Dy. DARPA-ACTM, *Neural Network Augmented Regional Physical Models and Koopman Operator Assisted Climate Tipping-point Modeling*. **Role Co-PI**. Grant Amount: \$1M.
5. D. Erdoğan, L. Su, E. Tunik, M. Yarossi, T. Imbiriba. *EMG Interfaces for All: Inclusive Feature Extraction and Privacy Preserving Intent Inference Across Diverse Populations*. **Role: co-PI**. Grant Amount: \$150K.
6. D. Erdoğan, A. Ramezani, G. Schirner, T. Imbiriba, *Cerebellum-inspired Self-learning Mixed-signal Neurocomputing for Motor Control*. Pre-proposal. NSF-EFRI, **Role: Co-PI**. Grant Amount: \$2M.

DoD Funding and Research Dr. Imbiriba is also a Co-Investigator in three ARL-funded projects part of *Foundational Research for Electronic Warfare in Multi-Domain Operations* (FREEDOM) program, involving the Northeastern University, Kosta's research institute, and Army Research Labs (ARL). The projects are:

1. BARMADS: Behavioral Analysis, Recognition, and Modelling for Adversarial Distributed Systems.
2. Interference Resilient Distributed Radar Waveform Design.
3. Radar networks for target recognition.

STUDENT SUPERVISION

PhD Students Mentoring

Help mentoring 14 Ph.D. students (5 female).

2019–Present Northeastern University

1. Ashutosh Singh, *Ph.D. Candidate*. Advisor: Deniz Erdoğan.
2. Ahmet Demirkaya, *Ph.D. Candidate*. Advisor: Deniz Erdoğan.
3. Beyza Kalkanli, *Ph.D. Candidate*. Advisor: Deniz Erdoğan.
4. Elifnur Sünger, *Ph.D. Candidate*. Advisor: Deniz Erdoğan.
5. Basak Celik, *Ph.D. Candidate*. Advisor: Deniz Erdoğan.
6. Kyle Lockwood, *Ph.D. Candidate*. Advisor: Deniz Erdoğan.
7. Yunus Bicer, *Ph.D. Candidate*. Advisor: Deniz Erdoğan.
8. Razieh Faghihpourayesh, *Ph.D. Candidate*. Advisor: Deniz Erdoğan.
9. Paul Ghanem, *Ph.D. Candidate*. Advisor: Deniz Erdoğan.
10. Niklas Smedemark-Margulies, *Ph.D. Candidate*. Advisor: Deniz Erdoğan.
11. Haoqing Li, *Ph.D. Candidate*. Advisor: Pau Closas.
12. Peng Wu, *Ph.D. Candidate*. Advisor: Pau Closas.
13. Gerald LaMountain, *Ph.D. Candidate*. Advisor: Pau Closas.
14. Helena Calatrava, *Ph.D. Candidate*. Advisor: Pau Closas.

2018–2021 Federal University of Santa Catarina

1. Ricardo Augusto Borsoi. *Ph.D. Candidate*. Advisor: José C. M. Bermudez.

Ph.D. Thesis Committee

2023 Northeastern University

1. Kyle Lockwood. Advisor: Deniz Erdoğan.
2. Yunus Bicer. Advisor: Deniz

2022 Northeastern University

1. Ashutosh Singh. Advisor: Deniz Erdoğan.
2. Elifnur Sünger. Advisor: Deniz Erdoğan.
3. Ahmet Demirkaya. Advisor: Deniz Erdoğan.
4. Beyza Kalkanli. Advisor: Deniz Erdoğan.
5. Guillermo Hernandez. Advisor: Pau Closas.
6. Peng Wu. Advisor: Pau Closas.

Undergraduate Thesis Committees

1. Carlos Patrick Alves da Silva. *Sistemas de Reconhecimento de Voz para o Português Brasileiro Utilizando os Corpora Spoltech e OGI-22*. 2008. Universidade Federal do Pará.
2. Gustavo Cunha Guedes. *Software em Java para Treinamento de Sistemas de Reconhecimento de Voz Baseados em Cadeias de Markov Escondidas (HMM)*. 2007. Universidade Federal do Pará.

PROFESSIONAL SERVICE

Session chair:

1. Chair of the “Pattern Recognition and Classification I” session, IEEE ICASSP 2023;

Reviewer for journals:

1. Journal of Applied Remote Sensing;
2. IEEE Transactions on Image Processing;
3. IEEE Transactions on Geoscience and Remote Sensing;
4. IEEE Access;
5. MDPI Sensors;
6. IEEE Signal Processing Letters;
7. IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP);
8. IEEE Transactions on Computational Imaging;
9. IEEE Transactions on Vehicular Technology.

PUBLICATIONS

Summary: 25 journal publications, 31 peer-reviewed conference/workshop publications, 3 abstracts.

Citations: 956, **h-index:** 16, **i10-index:** 21 (source: [Google Scholar](#)).

* indicates a student co-author I advised or mentored.

Journal Publications

- [1] T. Imbiriba, A. Demirkaya*, A. Singh*, D. Erdoğmus, and M. Goodwin. "Wearable Biosensing to Predict Imminent Aggressive Behavior in Psychiatric Inpatient Youths with Autism". In: *JAMA Network Open* (**accepted!**) (2023).
- [2] P. Wu*, T. Imbiriba, V. Elvira, and P. Closas. "Bayesian data fusion with shared priors". In: *IEEE Transactions on Signal Processing* (**accepted!**) (2023).
- [3] T. Imbiriba, O. Straka, J. Duník, and P. Closas. "Augmented physics-based machine learning for navigation and tracking". In: *IEEE Transactions on Aerospace and Electronic Systems* (2023), pp. 1–13. doi: [10.1109/TAES.2023.3328853](#).
- [4] R. A. Borsoi, D. Erdoğmuş, and T. Imbiriba. "Learning Interpretable Deep Disentangled Neural Networks for Hyperspectral Unmixing". In: *IEEE Transactions on Computational Imaging* 9 (2023), pp. 977–991. doi: [10.1109/TCI.2023.3321985](#).
- [5] R. A. Borsoi, T. Imbiriba, and P. Closas. "Dynamical Hyperspectral Unmixing With Variational Recurrent Neural Networks". In: *IEEE Transactions on Image Processing* 32 (2023), pp. 2279–2294. doi: [10.1109/TIP.2023.3266660](#).
- [6] H. Li*, B. Duvvuri, R. Borsoi, T. Imbiriba, E. Beighley, D. Erdoğmuş, and P. Closas. "Online fusion of multi-resolution multispectral images with weakly supervised temporal dynamics". In: *ISPRS Journal of Photogrammetry and Remote Sensing* 196 (2023), pp. 471–489. issn: 0924-2716. doi: <https://doi.org/10.1016/j.isprsjprs.2023.01.012>.
- [7] G. Stratis, P. Weinberg, T. Imbiriba, P. Closas, and A. E. Feiguin. "Sample generation for the spin-fermion model using neural networks". In: *Phys. Rev. B* 106 (20 2022), p. 205112. doi: [10.1103/PhysRevB.106.205112](#).
- [8] B. Azari, H. Cheng, N. Soltani, H. Li*, Y. Li, M. Belgiovine, T. Imbiriba, S. D'Oro, T. Melodia, Y. Wang, et al. "Automated deep learning-based wide-band receiver". In: *Computer Networks* (2022), p. 109367. doi: <https://doi.org/10.1016/j.comnet.2022.109367>.
- [9] J. Park, J. Moon, T. Kim, P. Wu*, T. Imbiriba, P. Closas, and S. Kim. "Federated Learning for Indoor Localization via Model Reliability With Dropout". In: *IEEE Communications Letters* 26.7 (2022), pp. 1553–1557. doi: [10.1109/LCOMM.2022.3170878](#).
- [10] N. Soltani, H. Cheng, M. Belgiovine, Y. Li, H. Li*, B. Azari, S. D'Oro, T. Imbiriba, P. Closas T. Melodia, Y. Wang, D. Erdoğmuş, and K. Chowdhury. "Neural Network-based OFDM Receiver for Resource Constrained IoT Devices". In: *IEEE Internet of Things Magazine* (Submitted) (2022). doi: <https://doi.org/10.48550/arXiv.2205.06159>.
- [11] H. Li*, R. A. Borsoi*, T. Imbiriba, P. Closas, J. C. M. Bermudez, and D. Erdoğmuş. "Model-Based Deep Autoencoder Networks for Nonlinear Hyperspectral Unmixing". In: *IEEE Geoscience and Remote Sensing Letters* 19 (2022), pp. 1–5. doi: [10.1109/LGRS.2021.3075138](#).
- [12] R. A. Borsoi*, T. Imbiriba, J.C.M. Bermudez, and C. Richard. "Fast Unmixing and Change Detection in Multitemporal Hyperspectral Data". In: *IEEE Transactions on Computational Imaging* 7 (2021), pp. 975–988. doi: [10.1109/TCI.2021.3112118](#).
- [13] R.A. Borsoi*, T. Imbiriba, J. C. M., C. Richard, J. Chanussot, L. Drumetz, J.-Y. Tournet, A. Zare, and C. Jutten. "Spectral Variability in Hyperspectral Data Unmixing: A comprehensive review". In: *IEEE Geoscience and Remote Sensing Magazine* 9.4 (2021), pp. 223–270. doi: [10.1109/MGRS.2021.3071158](#).
- [14] R. Faghihpirayesh*, M. Yarossi, T. Imbiriba, D. Brooks, E. Tunik, and D. Erdoğmuş. "Efficient TMS-based Motor Cortex Mapping Using Gaussian Process Active Learning". In: *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 29 (2021), pp. 1679–1689. doi: [10.1109/TNSRE.2021.3105644](#).
- [15] R. A. Borsoi*, T. Imbiriba, J. C. M. Bermudez, and C. Richard. "Deep Generative Models for Library Augmentation in Multiple Endmember Spectral Mixture Analysis". In: *IEEE Geoscience and Remote Sensing Letters* 18.10 (2020), pp. 1831–1835. doi: [10.1109/LGRS.2020.3007161](#).
- [16] R. A. Borsoi*, T. Imbiriba, P. Closas, J. C. M. Bermudez, and C. Richard. "Kalman Filtering and Expectation Maximization for Multitemporal Spectral Unmixing". In: *IEEE Geoscience and Remote Sensing Letters* 19 (2020), pp. 1–5. doi: [10.1109/LGRS.2020.3025781](#).

- [17] Ricardo Augusto Borsoi*, Tales Imbiriba, José Carlos Moreira Bermudez, and Cédric Richard. “A blind multiscale spatial regularization framework for kernel-based spectral unmixing”. In: *IEEE Transactions on Image Processing* 29 (2020), pp. 4965–4979. doi: [10.1109/TIP.2020.2978342](https://doi.org/10.1109/TIP.2020.2978342).
- [18] R. A. Borsoi*, T. Imbiriba, and J. C. M. Bermudez. “Deep Generative Endmember Modeling: An Application to Unsupervised Spectral Unmixing”. In: *IEEE Transactions on Computational Imaging* (2019), pp. 1–1. issn: 2573-0436. doi: [10.1109/TCI.2019.2948726](https://doi.org/10.1109/TCI.2019.2948726).
- [19] R. A. Borsoi*, T. Imbiriba, and J. C. M. Bermudez. “Super-Resolution for Hyperspectral and Multispectral Image Fusion Accounting for Seasonal Spectral Variability”. In: *IEEE Transactions on Image Processing* 29 (2020), pp. 116–127. issn: 1941-0042. doi: [10.1109/TIP.2019.2928895](https://doi.org/10.1109/TIP.2019.2928895).
- [20] T. Imbiriba, R. A. Borsoi*, and J. C. M. Bermudez. “Low-Rank Tensor Modeling for Hyperspectral Unmixing Accounting for Spectral Variability”. In: *IEEE Transactions on Geoscience and Remote Sensing* (2019), pp. 1–10. issn: 1558-0644. doi: [10.1109/TGRS.2019.2949543](https://doi.org/10.1109/TGRS.2019.2949543).
- [21] T. Imbiriba, J. C. M. Bermudez, J. Tournet, and N. J. Bershad. “A New Decision-Theory-Based Framework for Echo Canceled Control”. In: *IEEE Transactions on Signal Processing* 66.16 (2018), pp. 4304–4314. issn: 1053-587X. doi: [10.1109/TSP.2018.2849748](https://doi.org/10.1109/TSP.2018.2849748).
- [22] R. A. Borsoi*, T. Imbiriba, and J. C. M. Bermudez. “A Data Dependent Multiscale Model for Hyperspectral Unmixing With Spectral Variability”. In: *IEEE Transactions on Image Processing* (2020), pp. 1–1. issn: 1941-0042. doi: [10.1109/TIP.2020.2963959](https://doi.org/10.1109/TIP.2020.2963959).
- [23] R. A. Borsoi*, T. Imbiriba, J. C. M. Bermudez, and C. Richard. “A Fast Multiscale Spatial Regularization for Sparse Hyperspectral Unmixing”. In: *IEEE Geoscience and Remote Sensing Letters* 16.4 (2019), pp. 598–602. issn: 1558-0571. doi: [10.1109/LGRS.2018.2878394](https://doi.org/10.1109/LGRS.2018.2878394).
- [24] T. Imbiriba, J. C. M. Bermudez, and C. Richard. “Band Selection for Nonlinear Unmixing of Hyperspectral Images as a Maximal Clique Problem”. In: *IEEE Transactions on Image Processing* 26.5 (2017), pp. 2179–2191. issn: 1057-7149. doi: [10.1109/TIP.2017.2676344](https://doi.org/10.1109/TIP.2017.2676344).
- [25] T. Imbiriba, J. C. M. Bermudez, C. Richard, and J. Y. Tournet. “Nonparametric Detection of Nonlinearly Mixed Pixels and Endmember Estimation in Hyperspectral Images”. In: *IEEE Transactions on Image Processing* 25.3 (2016), pp. 1136–1151. issn: 1057-7149. doi: [10.1109/TIP.2015.2509258](https://doi.org/10.1109/TIP.2015.2509258).

Refereed Conference and Workshop Proceedings

- [1] N. Hanson, A. Demirkaya, D. Erdoğan, A. Stubbins, T. Padi, and T. Imbiriba. “A Vision for Cleaner Rivers: Harnessing Snapshot Hyperspectral Imaging to Detect Macro-Plastic Litter”. In: *IEEE WHISPERS 2023 - 13th Workshop on hyperspectral image and signal processing: evolution in remote sensing*. **Accepted!** 2023.
- [2] S. Asghari-Esfeden, G. Streng, K. Lockwood, Y. Bicer, T. Imbiriba, M. Furmanek, M. Yarossi, E. Tunik, T. Padi, and D. Erdogmus. “Real-Time Object Localization for Human-Robot Handover”. In: *Proceedings of the 16th International Conference on Pervasive Technologies Related to Assistive Environments*. PETRA '23. Corfu, Greece: Association for Computing Machinery, 2023, 42–46. isbn: 9798400700699. doi: [10.1145/3594806.3594854](https://doi.org/10.1145/3594806.3594854). url: <https://doi.org/10.1145/3594806.3594854>.
- [3] P. Wu, H. Calatrava, T. Imbiriba, and P. Closas. “Jammer classification with Federated Learning”. In: *2023 IEEE/ION Position, Location and Navigation Symposium (PLANS)*. **Best Technical Paper**. 2023, pp. 228–234. doi: [10.1109/PLANS53410.2023.10140124](https://doi.org/10.1109/PLANS53410.2023.10140124).
- [4] A. Nardin, T. Imbiriba, and P. Closas. “Crowdsourced Jammer Localization Using APBMs: Performance Analysis Considering Observations Disruption”. In: *2023 IEEE/ION Position, Location and Navigation Symposium (PLANS)*. 2023, pp. 511–519. doi: [10.1109/PLANS53410.2023.10140023](https://doi.org/10.1109/PLANS53410.2023.10140023).
- [5] S. Tang, G. LaMountain, T. Imbiriba, and P. Closas. “On Parametric Misspecified Bayesian Cramér-Rao Bound: An Application to Linear/Gaussian Systems”. In: *ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2023, pp. 1–5. doi: [10.1109/ICASSP49357.2023.10095758](https://doi.org/10.1109/ICASSP49357.2023.10095758).
- [6] A. Nardin, T. Imbiriba, and P. Closas. “Jamming Source Localization Using Augmented Physics-Based Model”. In: *ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2023, pp. 1–5. doi: [10.1109/ICASSP49357.2023.10095731](https://doi.org/10.1109/ICASSP49357.2023.10095731).
- [7] A. Singh*, A. Singh, A. Masoomi, T. Imbiriba, E. Learned-Miller, and D. Erdoğan. “Inv-Senet: Invariant Self Expression Network for Clustering Under Biased Data”. In: *ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2023, pp. 1–5. doi: [10.1109/ICASSP49357.2023.10094998](https://doi.org/10.1109/ICASSP49357.2023.10094998).
- [8] N. Smedemark-Margulies*, B. Celik, T. Imbiriba, A. Kocanaogullari, and D. Erdoğan. “Recursive Estimation of User Intent From Noninvasive Electroencephalography Using Discriminative Mod-

- els". In: *ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2023, pp. 1–5. doi: [10.1109/ICASSP49357.2023.10095715](https://doi.org/10.1109/ICASSP49357.2023.10095715).
- [9] R. A. Borsoi, T. Imbiriba, and D. Erdoğan. "A Deep Disentangled Approach for Interpretable Hyperspectral Unmixing". In: *ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2023, pp. 1–5. doi: [10.1109/ICASSP49357.2023.10095764](https://doi.org/10.1109/ICASSP49357.2023.10095764).
 - [10] K. Lockwood*, Y. Bicer, S. Asghari-Esfeden, T. Zhu, M. Furmanek, M. Mangalam, G. Streng, T. Imbiriba, M. Yarossi, T. Padi, D. Erdogmus, and E. Tunik. "Leveraging Submovements for Prediction and Trajectory Planning for Human-Robot Handover". In: *PETRA '22*. Corfu, Greece: Association for Computing Machinery, 2022, 247–253. ISBN: 9781450396318. doi: [10.1145/3529190.3529220](https://doi.org/10.1145/3529190.3529220).
 - [11] T. Imbiriba, A. Demirkaya*, J. Duník, O. Straka, D. Erdogmus, and Pau Closas. "Hybrid Neural Network Augmented Physics-based Models for Nonlinear Filtering". In: *2022 25th International Conference on Information Fusion (FUSION)*. 2022, pp. 1–6. doi: [10.23919/FUSION49751.2022.9841291](https://doi.org/10.23919/FUSION49751.2022.9841291).
 - [12] H. Li, B. Duvvuri, R. Borsoi*, T. Imbiriba, E. Beighley, D. Erdoğan, and P. Closas. "Online multi-resolution fusion of space-borne multispectral images". In: *2022 IEEE Aerospace Conference (AERO)*. IEEE. 2022, pp. 1–12. doi: [10.1109/AERO53065.2022.9843578](https://doi.org/10.1109/AERO53065.2022.9843578).
 - [13] A. Demirkaya*, T. Imbiriba, K. Lockwood*, S. Rampersad, E. Alhajjar, G. Guidoboni, Z. Danziger, and D. Erdoğan. "Cubature Kalman Filter Based Training of Hybrid Differential Equation Recurrent Neural Network Physiological Dynamic Models". In: *2021 43rd Annual International Conference of the IEEE Engineering in Medicine Biology Society (EMBC)*. 2021, pp. 763–766. doi: [10.1109/EMBC46164.2021.9631038](https://doi.org/10.1109/EMBC46164.2021.9631038).
 - [14] P. Wu*, T. Imbiriba, J. Park, S. Kim, and P. Closas. "Personalized Federated Learning over non-IID Data for Indoor Localization". In: *2021 IEEE 22nd International Workshop on Signal Processing Advances in Wireless Communications (SPAWC)*. 2021, pp. 421–425. doi: [10.1109/SPAWC51858.2021.9593115](https://doi.org/10.1109/SPAWC51858.2021.9593115).
 - [15] T. Imbiriba, P. Wu*, G. LaMountain, D. Erdoğan, and P. Closas. "Recursive Gaussian Processes and Fingerprinting for Indoor Navigation". In: *2020 IEEE/ION Position, Location and Navigation Symposium (PLANS)*. 2020, pp. 933–940. doi: [10.1109/PLANS46316.2020.9110187](https://doi.org/10.1109/PLANS46316.2020.9110187).
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